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Michael P. Draper, Hai-Yan Liu, Andreas H. Nelsbach, Stephen P. Mosley & Clyde L. Denis

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CCR4 Is a Glucose-Regulated Transcription Factor Whose Leucine-Rich Repeat Binds Several Proteins Important for Placing CCR4 in Its Proper Promoter Context†

MICHAEL P. DRAPER, HAI-YAN LIU, ANDREAS H. NELSBACH, STEPHEN P. MOSLEY, AND CLYDE L. DENIS*

Department of Biochemistry and Molecular Biology, University of New Hampshire, Durham, New Hampshire 03824

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The yeast CCR4 protein is required for the expression of a number of genes involved in nonfermentative growth, including glucose-repressible *ADH2*, and is the only known suppressor of mutations in the *SPT6* and *SPT10* genes, two genes which are believed to be involved in chromatin maintenance. We show here that although CCR4 did not bind DNA under the conditions tested, it was able to activate transcription when fused to a heterologous DNA-binding domain. The transcriptional activation ability of CCR4, in contrast to that of many other activators, was glucose regulated. Two activation domains one of which was glucose responsive and encompassed a glutamine-proline-rich region similar to that found in other eukaryotic transcriptional factors were identified. The two transactivation regions, when separated from the leucine-rich repeat and the C terminus of CCR4, were unable to complement a defective *ccr4* allele, suggesting that the leucine-rich repeat and the C terminus make contacts that link the activation regions to the proper gene context. Native immunoprecipitation of CCR4 revealed that CCR4 was complexed with at least four other proteins. The leucine-rich repeat of CCR4 was both necessary and sufficient for interaction with at least two of these factors. We propose that the leucine-rich repeat links CCR4 through its associated factors to its promoter context at *ADH2* and other loci where it is required for proper transcriptional regulation.

The general transcription factor CCR4 from *Saccharomyces cerevisiae* is required for the transcription of a number of genes involved in nonfermentative growth, including that of the *ADH2* gene (encoding the glucose-repressible alcohol dehydrogenase II) (11). Strains containing a deletion of *CCR4* also fail to grow at elevated temperatures on a nonfermentative medium, consistent with the global role played by CCR4 under these growth conditions (14). Cells lacking CCR4, however, display other phenotypes, suggesting that CCR4 is involved in processes in addition to that of controlling nonfermentative growth. *ccr4* mutations display a cold sensitivity phenotype under glucose growth conditions, a phenotype observed for defects in transcription initiation complex factors such as TFIIB, RPB1, SRB2, and SRB4 (1, 2, 29, 30, 33, 34, 37, 41). In addition, CCR4 is required for the elevated expression at the *ADH2* locus and for the altered transcriptional initiation at the *his4-912δ* locus that results from defects in the *SPT6* and *SPT10* genes (14). The *SPT6* and *SPT10* genes encode factors that are responsible for maintaining proper transcriptional control over a wide range of genes, and *SPT6* has been specifically implicated in the maintenance of chromatin structure (11, 26, 27, 39, 40).

spt6 mutations, moreover, suppress defects in the global transactivators SNF2 and SNF5, which are known to be involved in maintaining proper nucleosome positioning (5, 6, 19, 28, 43). Mutations in *SNF2* and *SNF5* affect the expression of many genes, cause defects in nonfermentative growth, and have been shown to be required for the activity of specific DNA-binding transcriptional activators (22). *CCR4*, the only known suppressor of *spt6*, may be functioning to counteract the repressive effects that SPT6 has on chromatin structure. CCR4 is also required for ADR1-dependent *ADH2* expression (11). ADR1 is a DNA-binding (16), gene-specific activator of *ADH2* (15). It is possible that, like the SNF2 and SNF5 proteins, CCR4 functions both in aiding gene-specific activators and in overcoming repression by nucleosomes (20, 22).

The CCR4 protein contains several regions which may be important to its function. The N-terminal region is rich in glutamines and asparagines, which is characteristic of a number of eukaryotic proteins involved in transcription (9, 25). The C terminus contains a region similar to the manganese binding site of xylose isomerases (unpublished data) (10, 20). The central region of the protein (residues 350 to 473) contains five tandem copies of a leucine-rich repeat. Leucine-rich repeat structures have been implicated in protein-protein interactions (3, 24, 38). Deletions within the CCR4 leucine-rich repeat abolish CCR4 function (25), suggesting that the interaction of the leucine-rich repeat with another protein is important to CCR4 activity. The possibility that CCR4 binds other proteins is consistent with several lines of evidence that suggest that general transcription factors function as components of large multisubunit complexes, such as observed for the SPT4, SPT5, and SPT6 proteins and the SWI1, SWI3, SNF2, and SNF5 proteins (32, 40).

We present evidence that CCR4 when bound to DNA via the LexA DNA-binding protein activates transcription in a glucose-responsive manner. Two activation domains, neither of which was capable of substituting for CCR4 function in vivo, were identified. CCR4 itself was unable to bind DNA, suggesting that it binds other proteins. CCR4 was found to be part of a multiprotein complex, and the leucine-rich repeat region was necessary and sufficient for CCR4 interaction with at least two components of this complex. These data suggest that the

* Corresponding author. Mailing address: Department of Biochemistry and Molecular Biology, University of New Hampshire, Durham, NH 03824. Phone: (603) 862-2427. Fax: (603) 862-4013.

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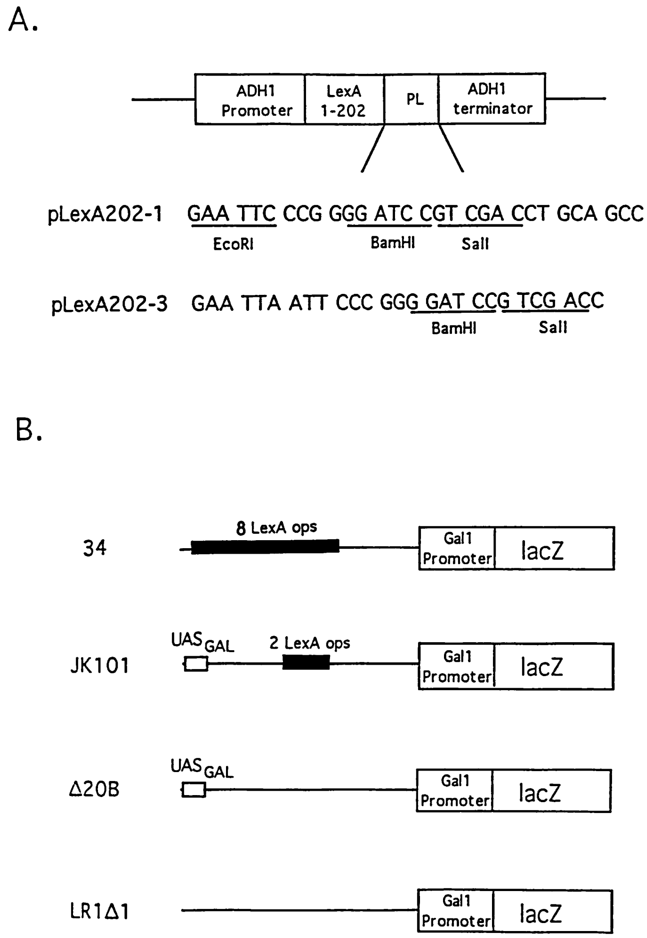

FIG. 1. Plasmids used in the study of transcriptional activation by LexA-CCR4 fusion proteins. (A) LexA expression plasmids used to produce LexA-CCR4 fusion proteins. Expression plasmids differed only in polylinker sequences (shown below partial plasmid diagram) into which CCR4 fragments were inserted. pLexA(1-87) is the same as pLexA202-1, except it contains only the N-terminal 87 residues of LexA. The proper reading frame is indicated by codon groups. Restriction enzyme sites that were used in the construction of LexA-CCR4 fusion plasmids are indicated. (B) LexA operator-controlled reporter plasmids. Two reporters were used to measure transcriptional activation by LexA-CCR4. Shown above is the 34 reporter containing eight LexA operators. The 1840 reporter (not shown) is identical to the 34 reporter, except that it contains a single LexA operator; the second and third plasmids were used to measure LexA-CCR4 binding to the LexA operator; the fourth plasmid was used as a control.

leucine-rich repeats make protein contacts which bring CCR4 to its proper promoter context.

MATERIALS AND METHODS

Plasmids. The LexA202 and LexA87 plasmids (Fig. 1A) are 2μ plasmids and have been previously described (4, 36). pLexA202-1 is the same plasmid as LexA(1-202)+pL (36). The 1840 and 34 reporter plasmids are 2μ-based plasmids with one and eight LexA operators, respectively, controlling the *lacZ* gene (4, 7). LexA-GAL4 contains residues 77 to 881 of GAL4 (22), and LexA-B42 contains an *Escherichia coli*-derived polypeptide that activates transcription in *S. cerevisiae* (36).

Plasmid constructions. The full-length LexA–CCR4(1-837) fusions were constructed by placing an *Eco*RI-*Bgl*II fragment from plasmid TM7 that contains residues 1 to 837 of CCR4 (25) into the *Eco*RI-*Bam*HI polylinker sites of the vectors pLexA202-1 and pLexA87-1. These LexA-CCR4 derivatives are designated pMD18 and pMD7, respectively (Fig. 1). The *Eco*RI site in each of these plasmids was made blunt with the large subunit of *E. coli* DNA polymerase (Klenow) to place *CCR4* coding sequences in frame with LexA. Deletion of amino acids 14 to 209 of CCR4 was accomplished by removing an *Apa*I fragment from the above-described two constructions. Truncation of CCR4 at amino acid 669 was conducted by filling in with Klenow the internal *Bam*HI site at bp 2004 of the LexA–CCR4(1-837) fusions, which results in four non-CCR4 amino acids RSKI at the C terminus. Construction of the LexA–leucine-rich repeat fusions was accomplished by digesting plasmid TM5 (pUC18 containing a *Hin*dIII [cuts at bp 472]-to-*Bam*HI [cuts at bp 2004] fragment of *CCR4*) with the *Taq*I enzyme and then filling in the resulting 5′ overhangs with Klenow. The *Taq*I fragments were subsequently cloned into pUC18, and after sequencing, one clone with an insertion containing bp 985 to 1421 of *CCR4* was selected for further manipulation. A *Bam*HI-*Eco*RI (filled with Klenow enzyme) fragment encompassing the leucine-rich repeats (residues 330 to 474) was then ligated into the *Bam*HI-*Sal*I (filled with Klenow) sites of both the pLexA87-3 and pLexA202-3 vectors. The *Bam*HI site was subsequently cut, filled in with Klenow, and religated to obtain the proper reading frame. The LexA fusion containing amino acids 404 to 837 of CCR4 was made by removing an *Eco*RI-*Bam*HI fragment from pMD18 and substituting the *Mun*I-*Bam*HI fragment from TM5 (see above description of TM5). The LexA-CCR4 fusions expressing amino acids 1 to 345 and 1 to 160 of CCR4 contain stop codons at bp 1033 and 478, respectively. The isolation of these fusions will be described elsewhere. LexA–CCR4-1-13/210-345 was made by removing an *Apa*I fragment from the plasmid containing LexA–CCR4-1-345. The LexA–CCR4-1-13 fusion plasmid was constructed by filling in with Klenow the *Bsp*120I site at bp 39 of the LexA–CCR4-1-837 plasmids and then religating. The LexA–CCR4-1-13 fusion has 58 non-CCR4 amino acids, PRTSGCTTSFAAHRFPAISHCRSLQITGRLHKASL DGNGRHSHRQQDDSKETTTYGR, at its C terminus.

Yeast strains, growth conditions, and enzyme assays. Yeast strains are listed in Table 1. Conditions for growth of cultures on minimal medium lacking uracil and histidine or YEP medium (2% Bacto Peptone, 1% yeast extract, 20 mg [each] of adenine and uracil [containing either 8% glucose or 2% ethanol as a carbon source] per liter) have been described (7). YD solid medium contained YEP supplemented to 2% glucose and 2.5% agar. β-Galactosidase assays were conducted on yeast extracts as described (7). Because yeast cells expressing LexA-CCR4 fusions proteins were observed to undergo loss of activity with prolonged maintenance on selective plates, assays were conducted within as short a time as possible on new transformants or freshly streaked-out colonies.

Native immunoprecipitations. Cells were grown in 2 liters of either YEP or an appropriate selective medium containing 4% glucose or 3% ethanol as described above. Cells were harvested at a density of 5×10^7 by centrifugation, were washed in 20 ml of ice-cold water, and were resuspended in 2 volumes of native extract buffer (25 mM KPO_4 buffer [pH 7.6], 150 mM KCl, 1 mM $NaPP_i$, 1 mM NaF, 5 mM $MgCl_2$, 1 mM EDTA, 10% glycerol, 1% Nonidet P-40, 8 μM leupeptin, 3 μM pepstatin A, 1 mM phenylmethylsulfonyl fluoride) prior to lysis in a bead beater according to the manufacturer's specifications. The resulting lysates were spun in a microcentrifuge for 15 min at maximum speed. The supernatant was then mixed with preimmune sera coupled to protein A-agarose at a 2:1 ratio (or

TABLE 1. Yeast strains used in this study

Strain	Genotype
237-1b	*MAT***a** *adh1-11 his3 leu2 trp1 ura3*
EGY188	*MAT***a** *his3 leu2 trp1 ura3 Lex*$_{OP}$*-LEU2*
MD9-7c	*MAT*α *adh1-11 his3 trp1 ura3 ccr4-10*
MD9-7c+	Same as MD9-7c except *ccr4-10::CCR4-TRP1*
876-2c	*MAT*α *adh1-11 his3 trp1 ura1/ura3 ccr4-391::HIS3 TRP1::ccr4-392/457*
906-6	*MAT***a** *adh1-11 his3 trp1 ccr4-391::HIS3 TRP1::ccr4-218/394*
86-6d	*MAT***a** *adh1-11 his3 trp1 ccr4-391::HIS3 TRP1::ccr4-1-13/210-837*
408-6d-TL5	*MAT*α *adh1-11 leu2 spt10-1 ccr4-1-669*
411-40	*MAT***a** *adh1-11 adh3 his4 trp1 ura1 adr1-1::1-ADR1-TRP1*
411-1	Same as 411-40 except *adr1-1::96-ADR1-TRP1*

protein A-agarose alone) and incubated at 4°C for 15 min with gentle rocking. After clarification in a clinical centrifuge, the precleared extract was mixed with antiserum coupled to protein A-agarose at a 5:1 ratio. The immune beads and the supernatant were incubated for 45 min at 4°C with rocking. Immune beads were washed three times with 10 volumes of ice-cold native extract buffer and twice in 10 volumes of ice-cold Triton wash buffer (20 mM $NaPO_4$ buffer [pH 7.6], 140 mM NaCl, 1 mM EDTA, 0.5% Triton X-100). The samples were then subjected to sodium dodecyl sulfate (SDS)-polyacrylamide gel electrophoresis (PAGE) and analyzed by Western immunoblotting or silver staining as previously described (42, 44).

DNA-binding assays. Gel retardation conditions, preparation of extracts, and incubation of extracts with DNA were as previously described (8). Gel mobility shift assays were conducted with a radiolabeled *Sau*3AI-*Eco*RV fragment of the *ADH2* promoter (bp −329 to +57). The DNA fragment was labeled with DNA polymerase I large fragment (Klenow) and with ^{32}P, which resulted in average activities of 30,000 cpm/ng. The DNA-binding mixture contained the following final concentrations: 2 μg of total protein, 100 mM KCl, 10% glycerol, 0.1 mM $ZnSO_4$, 0.1 μg of poly(dI-dC) per μl, and 0.1 ng of probe per ml in a 10-μl final volume. The mixture was incubated for 20 min on ice before being loaded onto either a 4 or a 6% polyacrylamide gel containing 2% glycerol and 2% Ficoll 400 that had been preelectrophoresed at 100 V for 60 min at 4°C in running buffer consisting of 190 mM glycine and 20 mM Tris, pH 8.0. The samples were subjected to electrophoresis at 400 V for 5 min and then 100 V for 3 h before the gel was dried and exposed for autoradiography. Extracts that were used in the incubations were from the following strains: MD9-7c (*ccr4*), MD9-7c+ (*CCR4*), and EGY188 containing the MD17 (LexA-CCR4) plasmid. Partially purified CCR4 was prepared from strain EGY188 that expressed a glutathione *S*-transferase–CCR4 fusion protein. Purification of glutathione *S*-transferase–CCR4 will be described elsewhere.

RESULTS

LexA-CCR4 transcriptional activation is carbon source regulated. The ability of CCR4 to activate transcription was monitored independently of the *ADH2* promoter context by the LexA in vivo transcription assay (4). The complete coding sequence for the yeast CCR4 protein was fused in frame with the coding sequence for the bacterial DNA-binding protein LexA (Fig. 1A). We expressed two versions of a LexA-CCR4 fusion protein, one containing only the DNA-binding domain of LexA (residues 1 to 87) and one containing the complete LexA protein (residues 1 to 202). Amino acid residues 88 to 202 of LexA contain a dimerization domain which is required for efficient DNA binding (17). The LexA-CCR4 fusion proteins expressed from these plasmids provided CCR4 function as assayed by complementation of the *ccr4-10* allele which causes a temperature-sensitive defect under nonfermentative growth conditions (Fig. 2). Also, both the 1 to 87 and 1 to 202 LexA-CCR4 fusion constructs complemented the inability of the strain carrying the *ccr4-10* allele to grow at 16°C on medium containing glucose (data not shown).

The LexA-CCR4 constructs were transformed into *S. cerevisiae* along with a *GAL1-lacZ* reporter gene under the control of the LexA operator (Fig. 1B). Both versions of the LexA-CCR4 fusion proteins activated expression of a *GAL1-lacZ* reporter gene containing either single or multiple LexA operators upstream of the GAL1 promoter (Table 2). However, LexA-CCR4 activated transcription to a much greater extent under nonfermentative growth conditions than under glucose-repressed conditions (Table 2). No activation by CCR4 was detected from the *GAL1-lacZ* target gene LR1Δ1 (Fig. 1) that lacked a LexA operator (data not shown), nor could the LexA DNA-binding domain alone activate expression from the reporters (Table 2). LexA-CCR4 protein concentration was found to be two- to three-fold lower under nonfermentative conditions than under glucose-repressed conditions (Fig. 3, compare lane b with lane a), suggesting that the ability of CCR4 to activate transcription in a carbon source-regulated manner was even greater than that reported in Table 2. While it is possible that the increased abundance of LexA-CCR4 under glucose growth conditions was titrating a factor required for its transcriptional activity, this seems unlikely. LexA-CCR4 activity would have been expected to increase dramatically in a strain lacking the genomic *CCR4* gene. However, this was not observed. LexA-CCR4 activity in a *CCR4* background was equal to or greater than that observed when LexA-CCR4 was placed in a *ccr4* strain background (data not shown). Western analysis also indicated that the LexA-CCR4 fusion proteins were present at only slightly elevated levels under glucose growth conditions compared with wild-type CCR4 levels and at lower levels than wild-type CCR4 under nonfermentative growth conditions (Fig. 3), suggesting that the ability of LexA-CCR4 to activate transcription was not due to an artificial overproduction of CCR4 protein. These results suggest that CCR4 when bound to DNA can function to activate transcription in a carbon source-regulated manner.

In contrast to CCR4, other known transcriptional activators fused to LexA did not display carbon source regulation. Several LexA-ADR1 derivatives, LexA-GAL4, and LexA-B42 (an *E. coli*-derived activator) all showed a reduction in transcriptional activity under nonfermentative growth conditions (Table 2) (7, 23, 36). It has also been previously reported that the activation potential of the general transcriptional factors SNF2 and SNF5, which like CCR4 are required for nonfer-

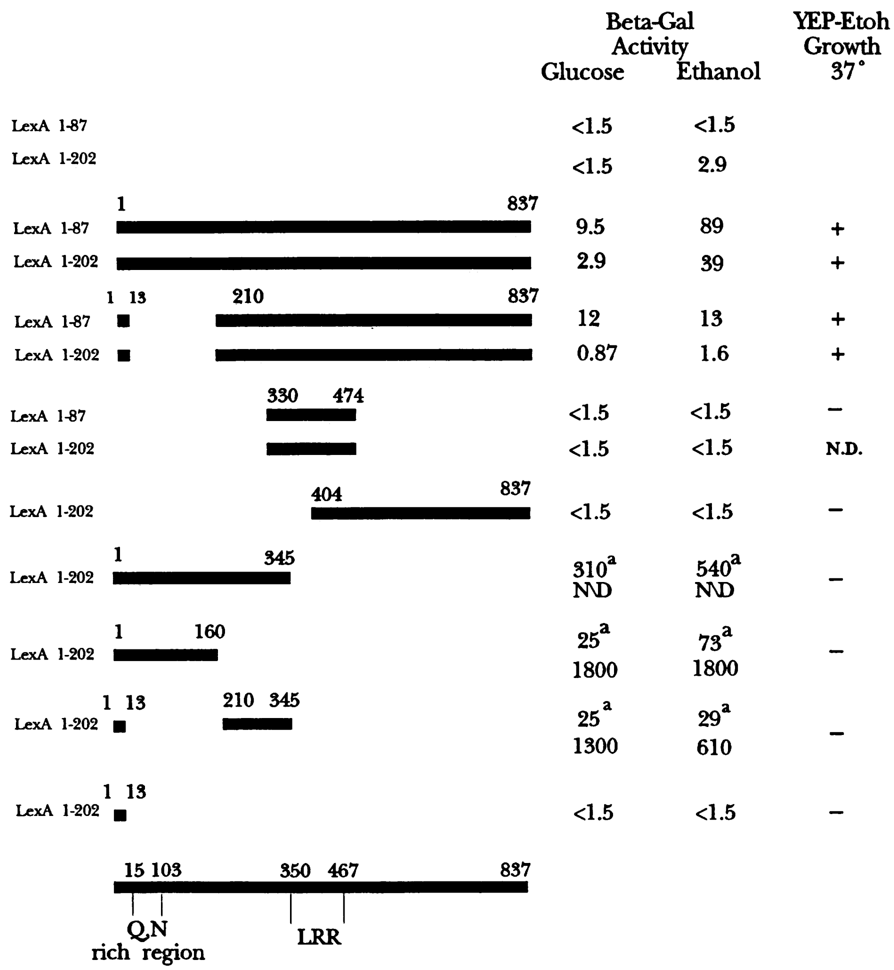

FIG. 2. Transcriptional activation by LexA-CCR4 fusion proteins. The solid bar represents CCR4 protein residues fused to either LexA(1-87) or LexA(1-202) as indicated. Strain 237-1b was grown in minimal medium lacking uracil and histidine and supplemented with either 8% glucose (Glucose) or 2% ethanol and 2% glycerol (Ethanol). All strains carried the 34 reporter plasmid, which has eight LexA binding sites, with the exception of the values for LexA–CCR4-1-345, LexA–CCR4-1-160, and LexA–CCR4-1-13/210-345, whose upper values were obtained from strains carrying the 1840 reporter (one LexA binding site). We were unable to obtain values for the 34 reporter with the LexA(1-202)–CCR4 (1-345) construct because of poor growth, possibly due to titration of a limiting transcription factor. β-Galactosidase (Beta-Gal) units are in units per milligram and represent the averages of at least three separate determinations. All standard errors of the mean were less than 20%, except for values of less than 20 U/mg, in which case standard errors were less than 30%. None of the LexA-CCR4 fusions was able to act with the LexA reporter LR1Δ1, which lacks LexA operator binding sites (Fig. 1). Western analysis indicated that all LexA-CCR4 fusions were of the expected size and of comparable abundance (data not shown). Complementation analysis was conducted with strain MD9-7c (*ccr4-10*), which displays temperature-sensitive growth at 37°C on nonfermentative medium. Similar results were obtained when the ability to complement the cold-sensitivity phenotype of *ccr4-10* (growth at 16°C on glucose-containing medium) was scored. [a], the 1840 reporter was used; ND, not determined.

mentative gene expression, were not glucose regulated when fused to LexA (23). The decrease in the ability of these other LexA fusion proteins reported in Table 2 to activate under nonfermentative growth conditions is likely due to their being under the control of the *ADH1* promoter (Fig. 1), which is derepressed on glucose and repressed under nonfermentative conditions (13). Western analysis confirmed a three- to fivefold reduction in the LexA-ADR1 proteins, a fivefold reduction in the LexA-B42 activator, and a threefold reduction in the amount of LexA-GAL4 present under nonfermentative growth conditions compared with glucose growth conditions (7; data not shown).

Many transcriptional factors contain sequence-specific DNA-binding domains. We therefore examined whether CCR4 could bind DNA by using a gel retardation assay used to identify ADR1 binding to *ADH2* sequences (8). We did not detect binding of CCR4 to *ADH2* regulatory sequences (bp −329 to +57) when using extracts from strains overexpressing CCR4 or extracts containing partially purified CCR4 (data not shown). Further, we saw no differences in the association of factors with *ADH2* regulatory sequences when extracts from *CCR4*$^+$- or *ccr4*-containing strains were compared (data not shown). Although the proper conditions for CCR4 binding to DNA may not have been uncovered, these results suggest that CCR4 neither binds DNA nor affects the binding of other factors to the *ADH2* promoter.

CCR4 contains two activation regions. The importance of the amino-terminal region of CCR4, which is rich in glutamines and prolines, to CCR4 function was examined by deleting amino acids 14 to 209. LexA(1-87)–CCR4-1-13/210-

TABLE 2. Transcriptional activity of LexA-CCR4 is glucose repressible

Fusion protein	No. of LexA operators[a]	β-Galactosidase activity (U/mg)	
		Glucose	Ethanol
LexA(1-87)–CCR4(1-837)	8	9	76
	1	4.8	33
LexA(1-202)–CCR4(1-837)	8	3.2	45
	1	2.9	15
LexA(1-87)	8	<1.5	<1.5
	1	<1.5	<1.5
LexA(1-202)	8	<1.5	2.9
	1	<1.5	<1.5
LexA(1-202)–ADR1(1-642)[b]	1	1,800	460
LexA(1-87)–ADR1(148-359)[b]	1	33	5.4
LexA(1-87)–B42	8	1,100	300
LexA(1-87)–GAL4	1	1,000	540

[a] LexA and all LexA fusions were expressed from a 2μm plasmid in strain 237-1b which contained either the 1840 reporter which has a single LexA operator controlling *lacZ* expression or the 34 reporter which contains eight Lex operator sites (Fig. 1B). Yeast cells were grown in minimal medium lacking uracil and histidine and which was supplemented with either 8% glucose (Glucose) or 2% glycerol and 2% ethanol (Ethanol). β-Galactosidase assays were conducted as described in Materials and Methods. All values represent the averages of at least three separate determination, and all standard errors of the means were less than 20%. Western analysis indicated that all LexA-CCR4 fusions were of the expected size and of comparable abundance (data not shown).

[b] Values were taken from reference 7.

837 was able to activate transcription of the *GAL1-lacZ* reporter gene under both fermentative and nonfermentative growth conditions, although its ability to activate under nonfermentative growth conditions was greatly diminished relative to that of full-length LexA-CCR4 (Fig. 2). This result implicates the 14 to 209 region as being responsible for the increased activity of LexA-CCR4 under nonfermentative conditions. It also suggests, however, that this is not the sole activation region present in CCR4. When the glutamine-rich region alone (residues 1 to 160) was fused to LexA, it functioned as an activator in a glucose-responsive manner, indicating that it did indeed contain an activation domain that was carbon source dependent. The glucose responsiveness of LexA–CCR4-1-160 was not apparent when a reporter containing eight LexA operator sites was used (Fig. 2). Deletion of residues 14 to 209 had no apparent effect, however, on the ability of LexA-CCR4 to complement a *ccr4-10* allele, again

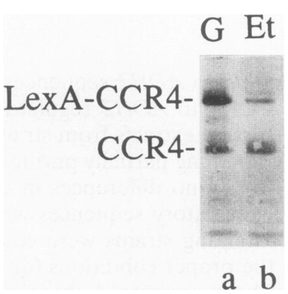

FIG. 3. Carbon source-dependent expression of LexA-CCR4. Strain 237-1b containing LexA(1-87)–CCR4-1-837 was grown in glucose- or ethanol-glycerol-containing medium as described in the legend to Fig. 2. Western analysis was conducted with crude antiserum raised against an N-terminal CCR4 peptide (25). G, glucose-grown cells; Et, ethanol-glycerol-grown cells. CCR4 and LexA-CCR4 proteins are indicated.

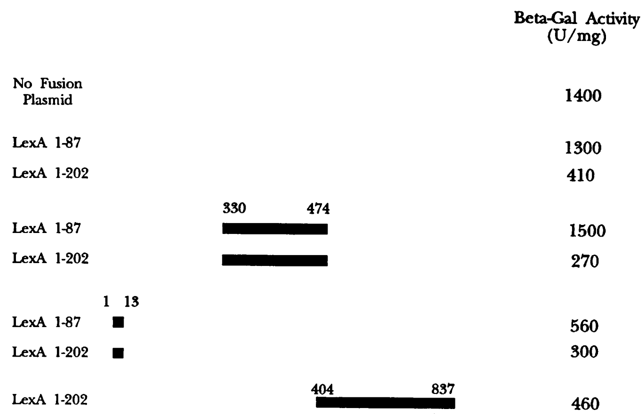

FIG. 4. Transcription interference assays with LexA-CCR4 derivatives. Yeast strains 237-1b and EGY188 containing the different LexA-CCR4 fusions and the JK101 reporter plasmid were grown in minimal medium lacking uracil and histidine and supplemented with 2% galactose and 3% glycerol. No substantive differences in values were observed between the two strains. Standard error of the mean values are <20%, except for those assay values under 100 U/mg whose standard error values were <25%. None of the LexA fusion proteins affected β-galactosidase (Beta-Gal) activity from plasmid Δ20B, which is the same as JK101 except that it lacks the LexA binding site (Fig. 1).

suggesting the presence of a second activation region in CCR4 (Fig. 2). A second transcriptional activation domain that was not glucose repressed was identified between residues 210 and 345, as demonstrated with LexA(1-202)–CCR4-1-13/210-345 (Fig. 2). Putting the two domains together [LexA(1-202)–CCR4-1-345] resulted in a synergistic increase in transcriptional activation.

In contrast, the leucine-rich repeat and the C terminus of CCR4 were incapable of activating transcription when fused to LexA (Fig. 2). In order to determine if the two LexA derivatives which lacked activation potential were still capable of binding the LexA operator, a transcription interference assay was used to monitor DNA binding of the fusion proteins. The interference assay utilizes the pJK101 plasmid which carries two LexA operators placed between the GAL4 upstream activation sequence (UAS) binding site and the *lacZ* reporter (Fig. 1B) (18). Under galactose growth conditions, GAL4 binds to its UAS site and activates transcription of the reporter. Binding of LexA fusion proteins to the LexA operator reduces this GAL4-induced transactivation. LexA fusions which were incapable of activation were examined with this assay. Both the LexA(1-202)–CCR4 derivatives containing only the leucine-rich repeats or the C terminus of CCR4 were capable of interfering with the GAL4-induced expression by at least threefold (Fig. 4). These data indicate that their inability to activate is not due to defects in binding to the LexA operator.

Each of the two activation domains was found to be much more active than the full-length LexA-CCR4 fusion (see Discussion). LexA–CCR4-1-345, LexA–CCR4-1-160, or LexA–CCR4-1-13/210-345 encompassing both or each activation domain failed, however, to complement the *ccr4-10* allele (Fig. 2). These results suggest that while residues 1 to 345 of the CCR4 protein are sufficient for transcriptional activation, it is likely that the leucine-rich repeats and the C terminus are required for placing these activation regions in the proper promoter context for CCR4 function.

CCR4 is required for ADR1 activation of transcription. The observation that CCR4 displays a carbon source-regulated

TABLE 3. Mutations in *CCR4* affect ADR1 activation ability

LexA fusion protein	CCR4 genotype	β-Galactosidase activity[a] (U/mg)	Fold decrease in activation
LexA(1-202)–ADR1(1-1323)	*CCR4*	1,300	2.4
	ccr4-10	540	
LexA(1-202)–ADR1(1-642)	*CCR4*	1,900	2.7
	ccr4-10	700	
LexA(1-202)–ADR1(359-740)	*CCR4*	780	11
	ccr4-10	74	
LexA(1-202)–ADR1(148-359)	*CCR4*	1,200	4.8
	ccr4-10	250	
LexA(1-202)–ADR1(1-220)	*CCR4*	2.8	5.6
	ccr4-10	<0.5	
LexA(1-87)–GAL4	*CCR4*	1,000	1.6
	ccr4-10	640	
LexA(1-87)–B42	*CCR4*	1,200	0.75
	ccr4-10	1,600	

[a] β-Galactosidase activities were conducted as described in footnote *a* to Table 2. The 1840 reporter was used in all cases except for LexA(1-202)–ADR1(148-359) and LexA(1-87)–B42, in which the 34 reporter containing eight LexA operator sites was used. All standard errors of the mean were less than 20%. Strains MD9-7c+ (*CCR4*) and MD9-7c (*ccr4-10*) are isogenic except for the *CCR4* allele.

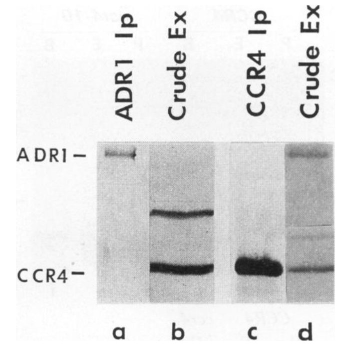

FIG. 5. ADR1 does not coimmunoprecipitate with CCR4. Western analysis was conducted with both purified ADR1 and purified CCR4 antibodies for all lanes. The band above the CCR4 protein is a nonspecific protein which the anti-CCR4 antibody recognizes in Western analysis. This protein is not immunoprecipitated, with this antibody, as shown in lane c. Lane a, ADR1-purified antibody to conduct the immunoprecipitation (Ip) from strain 411-40; lane b, yeast crude extract (crude Ex) from strain 411-40; lane c, CCR4-purified antibody to conduct the immunoprecipitation from strain 411-1; lane d, yeast crude extract from strain 411-1.

transcriptional activation ability when bound to DNA suggests that it functions as a coactivator with ADR1. We therefore examined the effect of a *ccr4* mutation on LexA-ADR1 activation of transcription. The *ccr4-10* allele had a 2.4-fold effect on the ability of the LexA–ADR1-1-1323 (full-length) protein fusion to activate. Since ADR1 contains three and possibly four functionally redundant activation regions, designated TADI through TADIV (7), we assessed whether LexA-ADR1 fusions containing fewer domains were also sensitive to the *ccr4-10* allele. As shown in Table 3, the *ccr4-10* allele also reduced by twofold the ability of LexA–ADR1-1-642 (contains three regions important to ADR1 activation function) to activate. LexA-ADR1 fusions which contained only one activation domain were more sensitive to a *ccr4-10* mutation. The activation ability of LexA–ADR1-359-740 (contains only TADIII), LexA–ADR1-148-359 (contains only TADII), and LexA–ADR1-1-220 (contains only TADI) were reduced 5- to 11-fold in the *ccr4-10* background. In comparison, the *ccr4-10* allele affected LexA–GAL4-74-881 (22)-mediated activation by 1.5-fold and had no effect on the *E. coli*-derived activator LexA-B42 (36). These results support the hypothesis that CCR4 plays a role in assisting ADR1 activation but also clearly indicate that CCR4 does not play an essential role.

ADR1 is not coimmunoprecipitated with CCR4. To determine if the requirement of CCR4 for ADR1 activation reflected a physical interaction between CCR4 and ADR1, we examined by immunoprecipitation the direct association of these two factors. ADR1 was immunoprecipitated from a native extract with antibody raised against an ADR1 C-terminal peptide. The precipitated proteins were subjected to Western analysis with an antibody raised against a CCR4 N-terminal synthetic peptide. ADR1 was specifically immunoprecipitated, as detected with anti-ADR1 antibody (Fig. 5, lane a) and failed to be precipitated when either preimmune sera or immune sera pretreated with excess antigen were used (42). CCR4 antibody, however, failed to detect CCR4 in the immunoprecipitates (Fig. 5, lane a), although CCR4 was present in the crude extract (Fig. 5, lane b). We repeated these experiments using CCR4 antibody to conduct the initial immunoprecipitation and anti-ADR1 antibody for detecting the presence of ADR1 following Western analysis. In this case, a strain carrying multiple copies of the *ADR1* gene was used to maximize the likelihood of detecting ADR1 in the immunoprecipitates. CCR4 was specifically precipitated by the anti-CCR4 peptide antibody (Fig. 5, lane c) but failed to be precipitated when preimmune sera or immune sera preincubated with excess peptide were used (25). ADR1, however, was not detected among the CCR4 immunoprecipitated proteins (Fig. 5, lane c), although it was present in the crude extract. These results indicate that CCR4 and ADR1 are not tightly associated in vivo.

Immunoprecipitation with CCR4 antibody reveals a multiprotein complex. Since the leucine-rich repeat is a putative protein binding domain and is required for CCR4 transcriptional function, we examined the possibility of a direct physical interaction between CCR4 and proteins other than ADR1 by determining which proteins coimmunoprecipitated with CCR4. Native extracts were treated with anti-CCR4 antibody, and the resultant immunoprecipitated proteins were subjected to SDS-PAGE and silver staining. Two abundant species were coimmunoprecipitated with CCR4 (Fig. 6A, lane b). These proteins had molecular masses of 195 and 185 kDa. Each was absent when preimmune sera or immune sera blocked with excess peptide was used (Fig. 6A, lane a and c, respectively). The presence of CCR4 in the immunoprecipitate was confirmed by Western analysis with the same samples (data not shown) and could also be visualized by silver staining (Fig. 6A, lane b). The 195- and 185-kDa species were not immunoprecipitated when native extracts of a strain lacking CCR4 (*ccr4-10* allele) were treated with anti-CCR4 antibody (Fig. 6A, lane e). The protein band in lane e in the 180- to 190-kDa region did not comigrate with the 185-kDa protein (Fig. 6B). These results indicated that CCR4 is associated with at least two other proteins to form a protein complex in vivo. Two additional proteins of molecular masses of 116 and 140 kDa

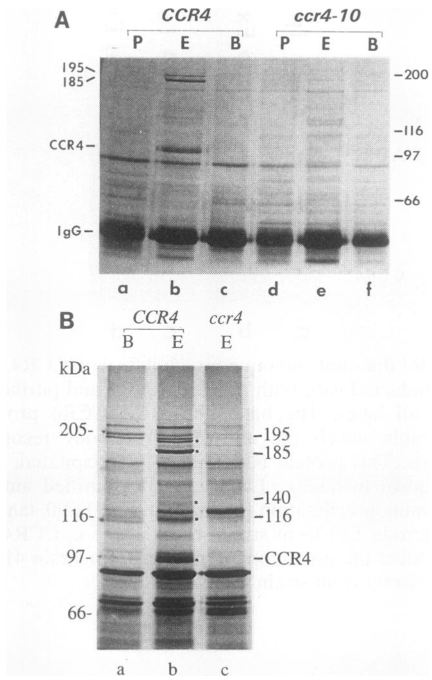

FIG. 6. CCR4 coimmunoprecipitates with several proteins. (A) Native immunoprecipitations were conducted, as described in Materials and Methods, under glucose growth conditions with strain MD9-7c+ (*CCR4*) or MD9-7c (*ccr4-10*). Molecular mass markers (in kilodaltons) are indicated on the right. Lanes a and d, incubation with preimmune (P) serum; lanes b and e, incubation with purified CCR4 antibody (E); lanes c and f, incubation with an excess of N-terminal CCR4 peptide prior to addition of CCR4 antibody (B). IgG, immunoglobulin G. (B) Immunoprecipitations were conducted as for panel A. The 195-, 185-, 140-, and 116-kDa and CCR4 proteins are indicated in lane b with black squares. Lane a, excess N-terminal CCR4 peptide was added prior to incubation of native extracts from strain MD9-7c+ (*CCR4*) with purified CCR4 antibody; lane b, same as lane a, except no peptide was added; lane c, same as lane b, except with strain MD9-7c (*ccr4-10*).

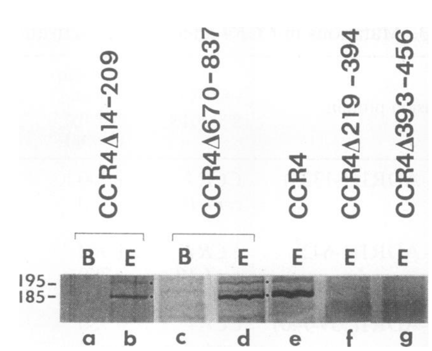

FIG. 7. The leucine-rich repeat is required for binding the 185- and 195-kDa proteins. Silver staining and immunoprecipitation of CCR4-associated proteins were conducted as described in the legend to Fig. 6. E, incubation with CCR4 antibody; B, incubation with excess peptide antigen prior to addition of CCR4 antibody. Lanes a and b, strain 86-6d (CCR4 1-13/210-837); lanes c and d, strain 408-6d-TL5 (CCR4-1-669); lane e, strain MD9-7c+ (CCR4); lane f, strain 906-6 (CCR4-1-218/395-837); lane g, strain 876-2c (CCR4-1-392/457-837). Other experiments indicated that the presence of the *spt10* allele in strain 408-6d-TL5 did not affect CCR4 protein expression (unpublished data).

were also observed to coimmunoprecipitate with CCR4 (Fig. 6B, lane b). The association of the 116- and 140-kDa polypeptides with CCR4 varied depending on the experiment, suggesting that they were less tightly associated with CCR4 than the 185- and 195-kDa proteins.

The leucine-rich repeats are necessary and sufficient for formation of the CCR4 protein complex. To identify which regions of CCR4 were required for binding the 195- and 185-kDa proteins, native extracts from strains containing alleles deleted for different regions of *CCR4* were treated with anti-CCR4 antibody. The immunoprecipitated proteins were treated as described above, and the 195- and 185-kDa species were detected by silver staining. As shown in Fig. 7, lanes b and d, the 195- and 185-kDa proteins were coimmunoprecipitated with CCR4 when residues 14 to 209 or 670 to 837 were removed from CCR4, respectively. The *ccr4-1-669* allele results in phenotypes indistinguishable from that of a *ccr4* deletion, and a LexA–CCR4-1-669 fusion was transcriptionally inactive (data not shown). When portions of the leucine-rich repeat region were deleted, however, the 195- and 185-kDa proteins were not coimmunoprecipitated (Fig. 7, lane f and g). The CCR4-Δ219-394 and CCR4-Δ393-456 proteins were present in the immunoprecipitates in an abundance comparable to that observed for the undeleted CCR4 proteins (25). These results indicate that the leucine-rich repeats are essential for the association of CCR4 with the 195- and 185-kDa proteins in vivo.

We subsequently examined the ability of the leucine-rich repeats by themselves to form a complex with the 195- and 185-kDa proteins. A portion of the *CCR4* gene containing the coding sequence for the leucine-rich repeats (residues 350 to 465) as well as a small amount of flanking sequences (total residues 330 to 474) were fused to the LexA(1-87) protein (designated LexA-LRR). The plasmid expressing the fusion protein was transformed into a yeast strain carrying a *ccr4-10* allele and into a strain carrying a wild-type *CCR4* gene. Immunoprecipitation of the fusion protein from native extracts of transformants expressing the LexA-LRR protein was carried out with an antibody raised against the LexA protein. The resultant proteins were subjected to SDS-PAGE and were visualized by silver staining. The 195- and 185-kDa proteins were coimmunoprecipitated in the presence of the LexA-LRR fusion protein (Fig. 8, lane a) but not from extracts that lacked the LexA-LRR (data not shown). This result demonstrates that the leucine-rich repeats are responsible for the interaction between CCR4 and the 195- and 185-kDa proteins. When wild-type CCR4 and LexA-LRR protein coexisted in the cell, the amount of 195- and 185-kDa proteins that coimmunoprecipitated with the LexA-LRR fusion protein was much less than that immunoprecipitated with LexA-LRR in a strain lacking CCR4 (Fig. 8, compare lane b with lane a). This indicates that LexA-LRR binds the 185- and 195-kDa proteins less well than full-length protein (also compare lane c with lane a).

Because CCR4 transcriptional activity is regulated by glucose (Table 2) and CCR4 is specifically required for nonfermentative gene expression, we wished to determine if the proteins associated with CCR4 did so in a carbon source-dependent manner. Native extracts from a wild-type strain of CCR4 following growth on glucose and ethanol were immunoprecipitated with anti-CCR4 antibody. Both the 195- and 185-kDa proteins were found to be coimmunoprecipitated with CCR4 when extracts taken under both growth conditions were used (Fig. 9, lanes a and b, respectively). The 140- and 116-kDa species that were shown to be coimmunoprecipitated with

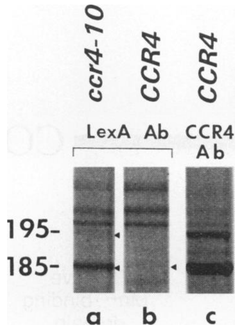

FIG. 8. The leucine-rich repeat binds the 185- and 195-kDa proteins. LexA-LRR expressed in strain MD9-7c or 237-1b was immunoprecipitated from nondenatured extracts with purified antibody directed against LexA or CCR4 N-terminal peptide as described in the legend to Fig. 6. Proteins were detected following SDS-PAGE by silver staining. As detected by Western analysis, LexA-LRR protein concentration was about twofold greater than that of CCR4 protein. Lane a, strain MD9-7c (*ccr4-10*) transformed with LexA-LRR and extracts treated with LexA antibody; lane b, strain 237-1b (*CCR4*) transformed with LexA-LRR and extracts treated with LexA antibody; lane c, strain 237-1b without LexA-LRR and extracts treated with CCR4 antibody.

CCR4 (Fig. 6B) were also complexed with CCR4 irrespective of the carbon source (data not shown). The decreased and varied abundance of the 140- and 116-kDa proteins in the immunoprecipitates precluded the determination of the specific region of CCR4 with which they interacted.

DISCUSSION

CCR4 displays a glucose-regulated transcriptional activation ability. Our results indicate that the yeast CCR4 protein is capable of activating transcription in a glucose-repressible manner when brought to the DNA through a heterologous DNA-binding protein. CCR4 itself, however, was not found to bind DNA. The ability of the LexA-CCR4 fusion proteins to function as activators correlated with their abilities to allow nonfermentative growth at elevated temperatures and to allow growth on glucose-containing medium at 16°C (Fig. 2) (25). For example, both leucine-rich repeat deletions and the C-terminal deletion of residues 670 to 837 inhibited CCR4 transcriptional activity as well as its ability to complement a

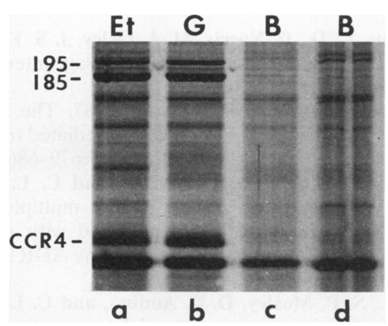

FIG. 9. Carbon source regulation of proteins binding to CCR4. The figure represents silver staining of proteins coimmunoprecipitating with CCR4, using antibody directed against the N-terminal CCR4 peptide. Cells from yeast strain MD9-7c+ (*CCR4*) were grown on YEP medium with the appropriate carbon source. G, glucose-grown cells; Et, ethanol-grown cells; B, excess N-terminal CCR4 peptide was added prior to addition of CCR4 antibody. Lanes b and c, glucose-grown cells; lanes a and d, ethanol-grown cells.

ccr4-10 mutation. These results indicate that LexA-CCR4 function in the LexA transcription assay mimics CCR4 function in vivo. The implication, therefore, is that there exists a mechanism by which CCR4 is brought to the vicinity of the DNA. This function is perhaps served by the factors which are complexed with CCR4. Whether CCR4 is a component of a transcriptional activator complex as observed for Hap 2, Hap 3, and Hap 4 in which there is a division of DNA-binding and activation functions is unclear (18, 31). It is equally possible that CCR4 plays a role modulating chromatin structure or the general transcriptional machinery.

The domain responsible for the glucose regulation of LexA-CCR4 was localized to the N-terminal region of CCR4. Deletion of residues 14 to 209 eliminated the derepression of CCR4 transcriptional activity upon glucose removal (Fig. 2), and residues 1 to 160 (designated TADI) by themselves displayed glucose-repressed activation (Fig. 2). This N-terminal region like many other eukaryotic transcription factors is rich in glutamines and prolines (Fig. 10). While it has been shown that the glutamine-rich activation domain of mammalian protein Sp1 does not function in *S. cerevisiae*, it is possible that the CCR4 glutamine-rich region is of a different type than that of Sp1 (35). Our data cannot distinguish between the CCR4 region being itself glucose regulatable or binding a protein whose activity is carbon source controlled.

A second transactivation domain was localized to residues 210 to 345 (designated TADII) that acted synergistically with TADI (Fig. 2). The transactivation displayed by TADI and TADII is in some ways different from that displayed by the full-length CCR4. First, the TADI and TADII domains are much more potent activators than full-length CCR4. It is possible that the TADI and TADII domains are more active by themselves because they have been released from an inhibition present in the full-length protein. Alternatively, the TADI and TADII domains may be displaying such potent activation because of their being released from the CCR4 protein complex. Full-length CCR4 transcriptional activation ability may be decreased because of its association with a number of factors that reduce its ability to activate. Second, the TADI and TADII domains show a multiplicative increase in activation potential as the number of LexA operators upstream of the LacZ reporter is increased. This is in contrast to what is observed with full-length CCR4, whose ability to activate increased only slightly with an increase in LexA operator sites. The C-terminal half of CCR4 may interact with a limiting factor, causing full-length CCR4 activity to increase only slightly with an increase in LexA operator sites. We believe that the transactivation ability of full-length CCR4 more accurately represents its true function rather than the high values observed for TADI and TADII alone.

It should also be noted that LexA(1-202)–CCR4 was less active than LexA(1-87)–CCR4 in activating transcription (Table 2; Fig. 2). One explanation for this difference is that the dimerization of LexA(1-202) interferes with CCR4 activity, perhaps by forcing an unusual conformation upon CCR4 or an unnatural arrangement of proteins associated with CCR4. This effect appears to be specific to CCR4, since a number of other proteins fused to LexA(1-202) do not display this diminution in activity (7, 36; unpublished data). This decreased activity of LexA(1-202)–CCR4 may account for the relatively lower activity observed for full-length CCR4 fused to LexA compared with that obtained with the LexA-CCR4 fusions containing just the N-terminal sequence of CCR4.

The leucine-rich repeat binds two proteins that may be required for presenting CCR4 to its proper promoter context. The leucine-rich repeats and the C terminus of CCR4 are

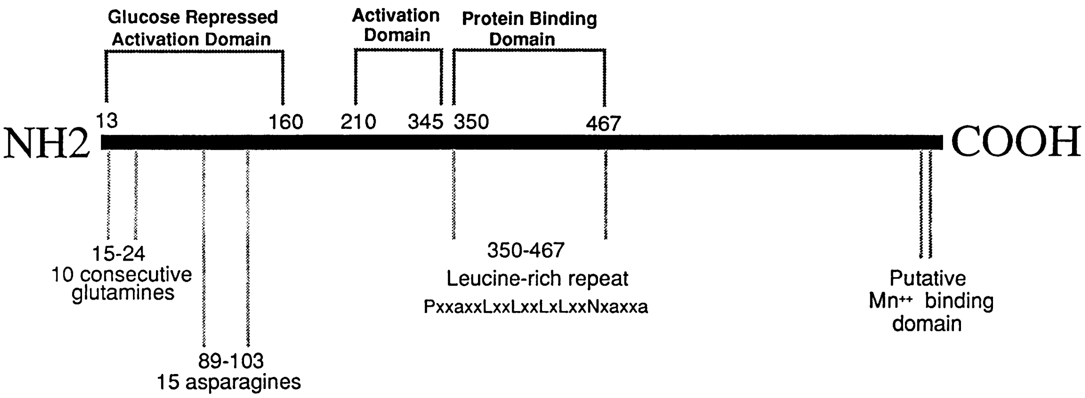

FIG. 10. CCR4 functional domains. CCR4 functional regions are indicated and described in the text.

required for CCR4 function, but neither region alone could promote transcription. The N-terminal domains described above function in that role, although these regions alone could not complement a *ccr4* allele in vivo. While it is possible that deletion of the leucine-rich repeats or C terminus inactivated CCR4 by placing the protein into an improperly folded form, we prefer the alternative explanation that the role of the leucine-rich repeats and the C terminus of CCR4 is to present CCR4 to its proper place at the promoter and to possibly regulate the N-terminal activation domains. The identification of two proteins, 195 and 185 kDa in size, that form a stable complex with the leucine-rich repeats of CCR4 reinforces this notion. The binding of the 195- and 185-kDa proteins to the leucine-rich repeat may make the leucine-rich repeat essential for CCR4 function. Two other proteins, 140 and 116 kDa in size, were also found to associate with CCR4. The site of binding for these latter two was not determined because of their weaker association with CCR4.

The contact that these 195- and 185-kDa proteins and CCR4 make at the *ADH2* promoter remains unclear. CCR4 was shown not to complex with ADR1, although CCR4 was required for maximal expression of LexA-ADR1 fusion proteins. We interpret these data to indicate that CCR4 is not a direct intermediate in ADR1 function but rather modulates some feature of *ADH2* transcription which impinges on ADR1 activation. Since CCR4 is the only known suppressor of the *spt6* mutation, a gene presumed to be involved in regulating or maintaining chromatin structure, CCR4 may be required to antagonize the negative effects of SPT6 (and possibly SPT10) on chromatin. The 195- and 185-kDa proteins appear to be neither SPT6 nor SPT10 (12; unpublished data). It is also possible that CCR4 acts independently of chromatin to promote transcription, perhaps by aiding the action of factors in the initiation complex.

Although a number of leucine-rich repeat proteins have been identified (see discussion in reference 25), in only two cases have the proteins that bind to these regions been identified. The leucine-rich repeat of RNase inhibitor binds RNase (24), and the leucine-rich region of thyrotropin receptor (and its related family members) binds thyrotropin hormone (and the corresponding hormones) (3). Preliminary analysis of contacts between these leucine-rich repeats and their cognate binding proteins have yet to yield a general consensus or structure characteristic of these interactions. The crystal structure of RNase inhibitor indicates that leucine-rich repeats consist of a parallel β-sheet with βα loops, but the precise site of RNase binding is not known (21). Because CCR4 contains the fewest number of leucine-rich repeats among this family of proteins (25), identifying its interaction regions with the 185- and 195-kDa proteins and cloning the genes for these proteins should aid in analyzing the protein-protein interactions of this important structural motif.

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The first two authors contributed equally to the results in this paper.

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